9 The function $f(x) = \dfrac{e^x}{1-e^x}$ is defined on the domain $x \in \mathbb{R}$, $x \neq 0$.

(i) Find $f^{-1}(x)$. **[3]**

(ii) Write down the range of $f^{-1}(x)$. **[1]**

9(i)

9(ii)